Theory Building: Issues with Rational Choice and Spatial Voting Theory

POL 51: Scientific Study of Politics

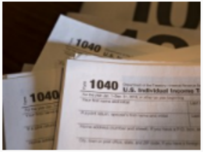
Spring 2026

Game theory in the real world. . .

Americans are proud to pay taxes — except when they think others are cheating.

If we fear others are getting a free ride, we become more likely to cheat.

Vanessa Williamson · **Public Opinion, Taxes** · 2 days ago



Review of rational choice theory

- **Homo economicus**: Humans are economic creatures.
- We weigh choices, and settle on the alternative that is likely to maximize utility.
- **Expected utility**: Probability of outcome $\times$ utility of outcome

Review of game theory

- Used to model situations in which multiple “players” seek to maximize their utility under conditions of uncertainty.
- Poker, international treaties, nuclear war, voting behavior, budget showdowns, etc.
- **Nash equillibrium**: an outcome in which players cannot unilaterally improve their position.
    - That is, given the opponent’s action, cannot do anything to gain more utility.
    - Doesn’t always exist, but does exist in prisoner’s dilemma and many other games.
- In prisoner’s dilemma games, such as with tragedy of the commons situations, the Nash equilibrium is not an optimal outcome for the players.

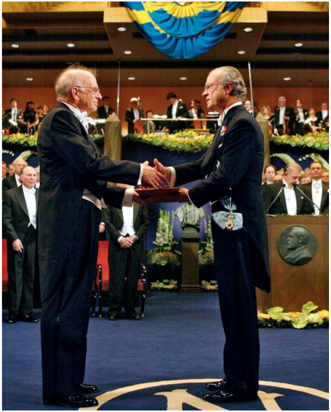
Problems?

- When does game theory fail (hint: ultimatum games)?
- Why do players behave irrationally?

Bounded Rationality

- Herbert Simon: Humans have cognitive limitations and are faced with a multitude of choices.
- **Bounded rationality**: We engage in "satisficing" by taking shortcuts (**heuristics**) to reach decisions.
- Example of heuristics: AAA Towing vs. ZZZ Towing?

Kahneman and Tversky

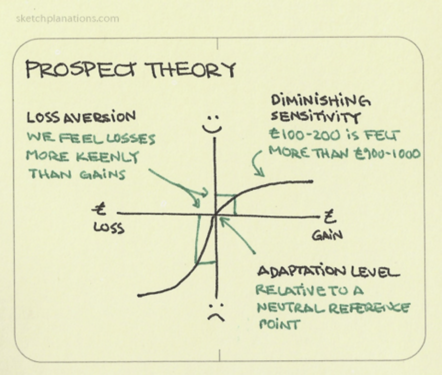


- Daniel Kahneman and Amos Tversky went further: humans have systematic biases that aren't just about information limits.
- Even in situations where we do have all relevant information about alternatives, we engage in behavior that does not maximize expected utility.
- Moreover, these deviations from expected behavior are *not* random, but systematic (predictable).

Anchoring

- **Anchoring**: Reference points matter. Starting values skew our preferences.

Prospect Theory



- Risk/loss aversion.

Example: Allais Paradox

Experiment 1				Experiment 2			
Gamble 1A		Gamble 1B		Gamble 2A		Gamble 2B	
Winnings	Chance	Winnings	Chance	Winnings	Chance	Winnings	Chance
$1 million	100%	$1 million	89%	Nothing	89%	Nothing	90%
		Nothing	1%	$1 million	11%		
		$5 million	10%			$5 million	10%

- Gamble 1A or Gamble 1B?
- Gamble 2A or Gamble 2B?

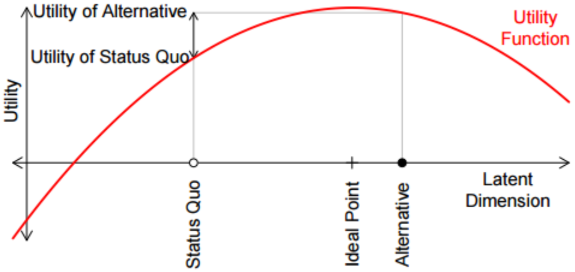
Example: Allais Paradox

Experiment 1				Experiment 2			
Gamble 1A		Gamble 1B		Gamble 2A		Gamble 2B	
Winnings	Chance	Winnings	Chance	Winnings	Chance	Winnings	Chance
$1 million	89%	$1 million	89%	Nothing	89%	Nothing	89%
$1 million	11%	Nothing	1%	$1 million	11%	Nothing	1%
		$5 million	10%			$5 million	10%

Spatial voting theory

How do we go from theory to specific (testable) hypotheses?

Spatial voting theory



- **Theory of spatial voting**: We can represent political actors, their preferences, and alternatives in abstract policy space. Actors will make choices that maximize expected utility.

Spatial voting theory

- **Median voter theorem**: In a single dimension with single-peaked preferences and an odd number of voters, the median voter always wins.
- More technically, outcomes converge to the ideal point of the median voter.
- Successes (pivots in Congress) and failures (party polarization).